

EX-NO:05**CASSANDRA CRUD OPERATIONS****3-11-2023****AIM:**

To Create a Column-Oriented database using Cassandra and manipulate the data.

CASSANDRA:

An open-source distributed NoSQL database management system designed to handle large amounts of data across many commodity servers without any single point of failure.

KEYSPACE:

```
cqlsh> create keyspace location
... with replication = {'class':'SimpleStrategy',
... 'replication_factor':1};
```

CRUD OPERATIONS:**CREATE COMMAND:**

```
cqlsh> use location;
cqlsh:location> |
```

```
cqlsh:location> create table location(
... Id int PRIMARY KEY,
... Name text,
... Age int,
... Loc_id int,
... Street_add text,
... City text,
... Country text );
```

INSERT COMMAND:

```
cqlsh:location> insert into location
... (Id, Name, Age, Loc_id, Street_add, City, Country)
... values
... (100, 'Steven', 20, 1000, '1297 Via Cola di Rie', 'Roma', 'Italy');
```

```
cqlsh:location> insert into location
... (Id, Name, Age, Loc_id, Street_add, City, Country)
... values
... (101, 'Neena', 22, 1100, '93091 Calle della Testa', 'Tokyo', 'Japan');
```

```
cqlsh:location> insert into location
... (Id, Name, Age, Loc_id, Street_add, City, Country)
... values
... (102, 'Bruce', 20, 1200, '2014 Jabberwocky Rd', 'Southlake', 'US');
```

```
cqlsh:location> insert into location
... (Id, Name, Age, Loc_id, Street_add, City, Country)
... values
... (103, 'Diana', 21, 1300, '2004 Charade Rd', 'Seattle', 'Cananda');
```

```
cqlsh:location> insert into location
... (Id, Name, Age, Loc_id, Street_add, City, Country)
... values
... (104, 'Ismael', 19, 1400, '1298 Vileparle', 'Bombay', 'India');
```

READ OPERATIONS:

```
cqlsh:location> select * from location;
```

id	age	city	country	loc_id	name	street_add
100	20	Roma	Italy	1000	Steven	1297 Via Cola di Rie
104	19	Bombay	India	1400	Ismael	1298 Vileparle
102	20	Southlake	US	1200	Bruce	2014 Jabberwocky Rd
101	22	Tokyo	Japan	1100	Neena	93091 Calle della Testa
103	21	Seattle	Cananda	1300	Diana	2004 Charade Rd

(5 rows)

ALTER COMMAND:

```
cqlsh:location> alter table location
... add Postal_code int;
cqlsh:location>
```

```
cqlsh:location> select * from location;
```

id	age	city	country	loc_id	name	postal_code	street_add
100	20	Roma	Italy	1000	Steven	null	1297 Via Cola di Rie
104	19	Bombay	India	1400	Ismael	null	1298 Vileparle
102	20	Southlake	US	1200	Bruce	null	2014 Jabberwocky Rd
101	22	Tokyo	Japan	1100	Neena	null	93091 Calle della Testa
103	21	Seattle	Cananda	1300	Diana	null	2004 Charade Rd

(5 rows)

INSERTING VALUES FOR POSTAL_CODE:

```
cqlsh:location> insert into location (id,postal_code) values
... (100,989);
cqlsh:location> insert into location (id,postal_code) values
... (102,26192);
cqlsh:location> insert into location (id,postal_code) values
... (103,98199);
cqlsh:location> insert into location (id,postal_code) values
... (101,1689);
cqlsh:location> insert into location (id,postal_code) values
... (104,490232);
```

DISPLAY THE TABLE AFTER INSERTING:

```
cqlsh:location> select * from location;
```

id	age	city	country	loc_id	name	postal_code	street_add
100	20	Roma	Italy	1000	Steven	989	1297 Via Cola di Rie
104	19	Bombay	India	1400	Ismael	490232	1298 Vileparle
102	20	Southlake	US	1200	Bruce	26192	2014 Jabberwocky Rd
101	22	Tokyo	Japan	1100	Neena	1689	93091 Calle della Testa
103	21	Seattle	Canada	1300	Diana	98199	2004 Charade Rd

```
(5 rows)
```

WHERE CLAUSE:

```
cqlsh:location> select * from location where age=21 allow filtering;
```

id	age	city	country	loc_id	name	postal_code	street_add
103	21	Seattle	Canada	1300	Diana	98199	2004 Charade Rd

```
(1 rows)
```

UPDATE OPERATIONS:

```
cqlsh:location> update location set name='Nancy', age=19
... where id=103;
cqlsh:location> |
```

DISPLAY TABLE AFTER UPADTING:

```
cqlsh:location> select * from location;
```

id	age	city	country	loc_id	name	postal_code	street_add
100	20	Roma	Italy	1000	Steven	989	1297 Via Cola di Rie
104	19	Bombay	India	1400	Ismael	490232	1298 Vileparle
102	20	Southlake	US	1200	Bruce	26192	2014 Jabberwocky Rd
101	22	Tokyo	Japan	1100	Neena	1689	93091 Calle della Testa
103	19	Seattle	Canada	1300	Nancy	98199	2004 Charade Rd

```
(5 rows)
```

DELETE OPERATIONS:

```
cqlsh:location> delete loc_id from location
... where id=100;
```

DISPLAY TABLE AFTER DELETING:

```
cqlsh:location> select * from location;
```

id	age	city	country	loc_id	name	postal_code	street_add
100	20	Roma	Italy	null	Steven	989	1297 Via Cola di Rie
104	19	Bombay	India	1400	Ismael	490232	1298 Vileparle
102	20	Southlake	US	1200	Bruce	26192	2014 Jabberwocky Rd
101	22	Tokyo	Japan	1100	Neena	1689	93091 Calle della Testa
103	19	Seattle	Cananda	1300	Nancy	98199	2004 Charade Rd

(5 rows)

CASSANDRA BATCH COMMAND:

```
cqlsh:location> begin batch
... insert into location (id, name, street_add, city, country)
... values
... (112, 'Jose', '12-98 Victoria Street', 'Sydney', 'Australia')
... update location set age=26, postal_code=2901
... where id=112;
... update location set loc_id=1000
... where id=100;
... apply batch;
```

DISPLAY TABLE AFTER USING BATCH COMMAND:

```
cqlsh:location> select * from location;
```

id	age	city	country	loc_id	name	postal_code	street_add
112	26	Sydney	Australia	null	Jose	2901	12-98 Victoria Street
100	20	Roma	Italy	1000	Steven	989	1297 Via Cola di Rie
104	19	Bombay	India	1400	Ismael	490232	1298 Vileparle
102	20	Southlake	US	1200	Bruce	26192	2014 Jabberwocky Rd
101	22	Tokyo	Japan	1100	Neena	1689	93091 Calle della Testa
103	19	Seattle	Cananda	1300	Nancy	98199	2004 Charade Rd

(6 rows)

CASSANDRA AGGREGATE FUNCTIONS:**COUNT COMMAND:**

```
cqlsh:location> select count(*) from location;
```

```
count
-----
      6
```

(1 rows)

SUM COMMAND:

```
cqlsh:location> select sum(postal_code) from location;

system.sum(postal_code)
-----
                620202

(1 rows)
```

SUM COMMAND USING WHERE:

```
cqlsh:location> select sum(postal_code) from location where age > 21 allow filtering;

system.sum(postal_code)
-----
                4590

(1 rows)
```

AVG COMMAND:

```
cqlsh:location> select avg(age) from location;

system.avg(age)
-----
                21

(1 rows)
```

MAX COMMAND:

```
cqlsh:location> select max(age) from location;

system.max(age)
-----
                26

(1 rows)
```

DISPLAY SOME COLUMNS THROUGH MAX COMMAND:

```
cqlsh:location> select id, name, Age, City, country, max(age) from location;

id | name | age | city | country | system.max(age)
---+---+---+---+---+---
112 | Jose | 26 | Sydney | Australia | 26

(1 rows)
```

MIN COMMAND:

```
cqlsh:location> select min(age) from location;

system.min(age)
-----
                19

(1 rows)
```

USING TTL (TIME-TO-LIVE):

```
cqlsh:location> create table loc(
... id int primary key,
... name text)
... with default_time_to_live=180;
```

INSERTING VALUES:

```
cqlsh:location> insert into loc(id, name)
... values
... (1, 'Japan');
cqlsh:location> insert into loc(id, name)
... values
... (2, 'russia');
cqlsh:location> insert into loc(id, name)
... values
... (3, 'india');
```

```
cqlsh:location> select * from loc;

id | name
---+-----
  1 | Japan
  2 | russia
  3 | india

(3 rows)
```

AFTER 180 SECONDS:

```
cqlsh:location> select * from loc;

id | name
---+-----

(0 rows)
```

CASSANDRA USER DEFINED DATA TYPES:

```
cqlsh:location> create type address(
    ... street text,
    ... city text,
    ... country text);
cqlsh:location> |
```

```
cqlsh:location> create table loc(
    ... id int primary key,
    ... name text,
    ... place address);
cqlsh:location> |
```

INSERTING VALUES:

```
cqlsh:location> insert into loc(id, name, place)
    ... values
    ... (100, 'Nancy',{street:'147 Spadina Ave', city:'Toronto', Country:'Canada'}});
cqlsh:location> insert into loc(id, name, place)
    ... values
    ... (101, 'Jose',{street:'12-98 Victoria Street', City:'Sydney', Country:'Australia'}});
cqlsh:location> insert into loc(id, name, place)
    ... values
    ... (102, 'Luis',{street:'198 Clement North', City:'Singapore', Country:'Singapore'}});
cqlsh:location> |
```

DISPLAY THE VALUES:

```
cqlsh:location> select * from loc;

 id | name | place
-----+-----+-----
 100 | Nancy | {street: '147 Spadina Ave', city: 'Toronto', country: 'Canada'}
 102 | Luis  | {street: '198 Clement North', city: 'Singapore', country: 'Singapore'}
 101 | Jose  | {street: '12-98 Victoria Street', city: 'Sydney', country: 'Australia'}

(3 rows)
cqlsh:location> |
```

RESULT:

Thus the column-oriented database Cassandra have been created and all the basic crud operations and some more commands are executed successfully.

OBSERVATION	RECORD	SIGNATURE